# César Antonio Carvajal de la Haza

Mathematics · Statistics · Physics | M.Sc. student in Quantum Technologies

Seville, Spain · cesardelahaza@hotmail.es · cesardelahaza.github.io · github.com/cesardelahaza · linkedin.com/in/cesardelahaza

## Profile

Mathematics, Statistics and Physics graduate, now doing an M.Sc. in Quantum Technologies (QTEP) alongside full-time work. Research experience in quantum many-body physics from a final degree project on entanglement entropy, and applied machine learning through end-to-end projects, with Python as main language. Mathematics is what I enjoy most, which is why quantum technologies appeal to me: they bring mathematics, physics and computing together.

## Education

**M.Sc. in Quantum Technologies (QTEP)** Oct 2026 – present
Universidad Internacional Menéndez Pelayo (UIMP), Spain

- Courses in progress: fundamentals of quantum technologies, theory of quantum information, advanced quantum theory, quantum computing, machine learning and quantum computers, quantum cryptography and communication, quantum sensors (metrology and sensing), quantum technologies with photons and atoms, open quantum systems and quantum thermodynamics, and quantum technologies laboratory; Master's thesis.

**B.Sc. in Physics · GPA 8.1/10** 2021 – 2025
National University of Distance Education (UNED), Spain · 240 ECTS

- Honours in Classical Mechanics, Vibrations & Waves, Quantum Physics II and Dynamical Systems.
- Final degree project on entanglement entropy and geometry in quantum systems (see Projects).

**Double B.Sc. in Mathematics and Statistics · GPA 8.17/10** 2016 – 2021
University of Seville, Spain · 342 ECTS

- Mathematics: algebra, differential equations, topology, probability, coding theory and cryptography, computing. Programming in Haskell, Python, Maxima and MATLAB.
- Statistics: statistical inference, multivariate analysis, computational statistics, databases, operations research, AI & statistics. Projects in R.
- Honours in Survey Design; took part in the 2019/20 user satisfaction study of the University's IT service (survey design, data collection, analysis in R and Excel, final report).

## Projects

**Entanglement Entropy and Geometry in Quantum Systems** 2024 – 2025
Final degree project, Physics · graded 9.5/10 · GitHub · Outreach article (UNED Physics Blog, 2025)

- Investigated how interaction geometry shapes entanglement in hard-core boson systems, testing the entanglement Area Law.
- Modelled hopping with a Hubbard Hamiltonian on graphs (periodic chains, bridge and tree geometries); computed ground states by numerical diagonalisation and pairwise entanglement entropies from reduced density matrices.
- Built the Entanglement Adjacency Matrix (EAM) to compare physical connectivity with quantum correlations, identifying Area Law violations and entanglement monogamy effects. All simulations in Python.

**MLWeather: Temperature Forecasting with ML** 2026
Personal project · GitHub · Live app

- Built an end-to-end forecasting pipeline for Seville on hourly data from the Open-Meteo API.
- Compared Linear Regression, Random Forest, XGBoost, SARIMA and LSTM, evaluated with MAE, RMSE and $R^2$; used lag features, cyclical time encodings and a strict train/test split to avoid data leakage.
- Shipped an interactive Streamlit app with a modular architecture for exploring and comparing predictions. scikit-learn, XGBoost, Keras.

**GNSS + IMU Sensor Fusion Simulator** 2025
Capstone of the GNSS Academy six-month intensive programme (Jan – Jul 2025: DGNSS, RTK, PPP, SBAS, gLAB) · Python

- Simulated GNSS measurements with SP3 orbit propagation and realistic error models (ionospheric, tropospheric, clock, multipath), with Antenna Phase Offset and Sagnac corrections.
- Implemented IMU mechanisation equations in the ECEF frame and a least-squares module for GNSS-only positioning; fused GNSS and IMU data with a Kalman filter.

**Algorithmic Problem Solving** 2026 – present

Self-directed · GitHub

- LeetCode problems in C++ (arrays, linked lists, two pointers, binary search, dynamic programming) and SQL (joins, aggregations).

## Experience

**Consulting Analyst, RPA** Jun 2026 – present

Minsait (Indra Group), Data, Intelligence & Robots unit · Spain

- Robotic Process Automation; completed the UiPath Automation Developer Professional Training during onboarding.

**Secondary School Mathematics Teacher** Jan 2026 – Jun 2026

Colegio Internacional Alminar · Spain

- Taught Mathematics from 3º ESO to 2º Bachillerato (ages 14 to 18): algebra, calculus, statistics and geometry.
- Planned the curriculum, prepared exams and materials for each level, and designed personalised adaptations for students with diverse learning needs.

**Private Tutor** 2022 – 2025

Self-employed, and at the Refuerza study centre in Seville (Oct 2023 – Jun 2024)

- Mathematics, Physics, Chemistry and Technical Drawing, with a focus on clarity, logical reasoning and problem-solving.

**QA Automation Engineer** Apr 2022 – Nov 2022

Deloitte · Spain · official job title: Junior Delivery Analyst

- Developed automated test suites in Selenium and Java, reducing manual testing time.
- Co-implemented an automation framework with Selenium, Cucumber and Jenkins; tracked defects in Jira and worked with Git in an international team.

## Training

**Introduction to Quantum Computing · 1.5 ECTS** Mar 2026 – Apr 2026

UNED extension course

- Qubits, gates, superposition and entanglement; Grover's and Shor's algorithms; hands-on circuits in PennyLane and Qiskit.

## Skills

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|---|---|
| Programming | Python (main language), C/C++ (intermediate), SQL (intermediate), R, Haskell, MATLAB, Bash (basic) |
| Machine learning | scikit-learn, XGBoost, Keras (LSTM), Streamlit |
| Quantum | Quantum mechanics, entanglement entropy, quantum computing (introductory): PennyLane, Qiskit |
| Tools | Git, Jupyter, LaTeX, VSCode, gLAB, Linux command line (basic) |
| Languages | Spanish (native), English (C1, Aptis, British Council), French (basic, in progress) |

## Volunteering

- **El Buen Amigo animal shelter**: weekly care for rescued dogs; adoption advertisements and awareness campaigns.
- **Colectivo LGTB+ Pedro Zerolo**, Faculty of Mathematics, University of Seville (2017 – 2019): awareness campaigns and outreach activities.